

Part X - Rheumatology and Immunology

Chapter 10C: Lupus, Antiphospholipid Syndrome, Sjogren Syndrome, Systemic Sclerosis, Myositis, and Overlap Disease

This chapter is an original Medvale synthesis of connective tissue diseases that share systemic inflammation, autoantibody production, vascular injury, immune-complex activity, and multi-organ complications. It is designed for a custom medical study book and future Graph-RAG extraction, so the text emphasizes relationships among symptoms, antibodies, organ systems, emergency presentations, and treatments rather than isolated fact lists.

Core use pattern: Start with the syndrome. Photosensitive rash plus nephritis suggests SLE; thrombosis or pregnancy morbidity suggests antiphospholipid syndrome; dry eyes and dry mouth suggest Sjogren disease; Raynaud plus skin thickening suggests systemic sclerosis; proximal weakness plus CK elevation suggests inflammatory myopathy. Then search for red flags: nephritis, pulmonary hemorrhage, CNS disease, rapidly progressive interstitial lung disease, renal crisis, digital ischemia, pulmonary hypertension, dysphagia, myocarditis, or catastrophic thrombosis.

1. Connective tissue disease framework

Connective tissue diseases often begin with nonspecific symptoms: fatigue, low-grade fever, arthralgia, Raynaud phenomenon, puffy fingers, photosensitivity, oral ulcers, cytopenias, sicca symptoms, dysphagia, or dyspnea. Early disease may not satisfy formal classification criteria. The safest diagnostic approach is to define the phenotype, test for antibodies that match that phenotype, screen for silent organ involvement, and exclude mimics such as infection, malignancy, medication reactions, endocrine disease, viral arthritis, and noninflammatory pain syndromes.

A positive autoantibody rarely diagnoses disease by itself. ANA is sensitive for SLE but nonspecific. RF appears in Sjogren disease, chronic infection, and older adults. Antiphospholipid antibodies may be transient after infection. Myositis antibody panels help identify clinical subsets, but objective weakness, CK elevation, rash, ILD, biopsy, MRI, or EMG evidence may still be needed. The diagnosis is strongest when clinical syndrome, antibody pattern, and organ findings point in the same direction.

Presentation pattern	Likely disease cluster	Tests that change management
Photosensitive rash, oral ulcers, inflammatory arthralgia, cytopenias, proteinuria	Systemic lupus erythematosus	ANA, anti-dsDNA, anti-Sm, C3/C4, urinalysis with microscopy, urine protein/creatinine, creatinine, CBC
Venous/arterial thrombosis, recurrent pregnancy loss, livedo reticularis, false-positive syphilis test	Antiphospholipid syndrome	Lupus anticoagulant, anticardiolipin IgG/IgM, anti-beta-2 glycoprotein I IgG/IgM, repeated at least 12 weeks apart
Dry eyes, dry mouth, dental caries, parotid enlargement, neuropathy	Sjogren syndrome	Anti-Ro/SSA, anti-La/SSB, ANA/RF, Schirmer or ocular staining, salivary evaluation, lymphoma red flags
Raynaud, puffy fingers, skin thickening, reflux/dysphagia, telangiectasias, ILD, pulmonary hypertension	Systemic sclerosis	Anticentromere, anti-topoisomerase I/Scl-70, anti-RNA polymerase III; PFTs/DLCO, echocardiography, renal/BP monitoring
Symmetric proximal weakness, high CK, dysphagia, heliotrope rash, Gottron papules, mechanic hands, ILD	Idiopathic inflammatory myopathy	CK/aldolase, AST/ALT/LDH, myositis antibody panel, MRI/EMG, muscle biopsy when needed, ILD and malignancy screening
Raynaud, swollen hands, inflammatory arthritis, myositis, esophageal dysmotility, high anti-U1-RNP	Mixed connective tissue disease or overlap	ANA, anti-U1-RNP, CK, pulmonary evaluation, lupus/scleroderma/myositis organ screening

2. Systemic lupus erythematosus

Systemic lupus erythematosus (SLE) is a chronic multisystem autoimmune disease characterized by loss of tolerance to nuclear antigens, formation of autoantibodies, immune-complex deposition, complement activation, and tissue injury. It can affect skin, joints, kidneys, serosal surfaces, blood cells, lungs, heart, blood vessels, and the nervous system. The same diagnosis may look very different in different patients: one has photosensitive rash and arthralgia, another has proliferative nephritis, and another presents with seizure, thrombocytopenia, and antiphospholipid antibodies.

The major clinical task in suspected SLE is to identify organ-threatening disease. Rash, fatigue, and joint pain are common, but nephritis, CNS disease, pulmonary hemorrhage, myocarditis, severe cytopenias, mesenteric vasculitis, and severe thrombosis determine urgency. A patient with lupus symptoms should always have a blood pressure, CBC, creatinine, urinalysis with microscopy, and protein quantification when urine abnormalities are present.

2.1 SLE clinical manifestations

- **Constitutional:** fatigue, malaise, fever, and weight loss are common but nonspecific; infection and malignancy must be considered, especially in immunosuppressed patients.
- **Mucocutaneous:** malar rash sparing the nasolabial folds, photosensitivity, discoid lesions, subacute cutaneous lupus, oral/nasal ulcers, nonscarring alopecia, Raynaud phenomenon, livedo when APS coexists.
- **Musculoskeletal:** arthralgia and nonerosive inflammatory arthritis are common. Deforming but nonerosive Jaccoud arthropathy may occur; erosive arthritis suggests RA overlap or another diagnosis.
- **Serosal/cardiopulmonary:** pleuritis, pleural effusion, pericarditis, pericardial effusion, myocarditis, pneumonitis, ILD, pulmonary hypertension, shrinking lung syndrome, and rare diffuse alveolar hemorrhage.
- **Hematologic:** leukopenia, lymphopenia, thrombocytopenia, autoimmune hemolytic anemia, anemia of chronic inflammation, and cytopenias from medication or infection.
- **Renal:** proteinuria, hematuria, RBC casts, hypertension, nephrotic syndrome, or rising creatinine. Lupus nephritis may be silent early, so urinalysis surveillance is essential.
- **Neurologic:** seizures, psychosis, stroke, cognitive dysfunction, transverse myelitis, mononeuritis multiplex, headache syndromes. Exclude infection, metabolic causes, medications, and APS thrombosis.

SLE red flags: proteinuria, cellular casts, rising creatinine, hemoptysis, hypoxemia, pulmonary infiltrates, seizures, psychosis, focal neurologic deficits, severe thrombocytopenia, hemolytic anemia, myocarditis, mesenteric ischemia, or pregnancy with antiphospholipid antibodies should interrupt routine outpatient workup and prompt urgent specialty involvement.

2.2 SLE immunology and autoantibody interpretation

SLE immunology centers on abnormal handling of apoptotic nuclear material and immune sensing of nucleic acids. B cells form antibodies to nuclear antigens; immune complexes deposit in glomeruli, skin, small vessels, synovium, and serosal surfaces. Complement is consumed during immune-complex activation. Low C3 and C4 often support active lupus nephritis or systemic immune-complex activity, although complement levels do not perfectly track every manifestation.

SLE antibody/lab	Interpretation	Clinical associations and cautions
ANA	Sensitive screening test but not specific.	Almost all classic SLE patients have ANA positivity, but ANA also occurs in healthy people and other autoimmune diseases. A positive ANA alone does not diagnose lupus.
Anti-dsDNA	Specific for SLE and often activity-associated.	Rising anti-dsDNA plus falling complement and active urine sediment suggests lupus nephritis flare. Not all patients track with dsDNA.
Anti-Sm	Highly specific but insensitive.	Supports SLE when clinical features fit; not a general disease activity marker.
Anti-Ro/SSA	Seen in SLE, subacute cutaneous lupus, Sjogren disease, and neonatal lupus risk.	Important in pregnancy because maternal antibodies can cause neonatal lupus and congenital heart block. Can be present when ANA assay is negative.
Anti-La/SSB	Often paired with anti-Ro in Sjogren/overlap.	Interpret with sicca features and anti-Ro status; less useful as isolated test.
Antihistone	Common in drug-induced lupus.	Hydralazine, procainamide, isoniazid, minocycline, anti-TNF agents, and other drugs can cause lupus-like disease. Severe nephritis/CNS disease is less typical.
Antiphospholipid antibodies	Persistent antibodies plus clinical thrombosis or pregnancy morbidity define APS.	Associated with thrombosis, pregnancy loss, livedo, thrombocytopenia, valve disease, and false-positive RPR/VDRL.
Low C3/C4	Complement consumption from immune complexes or inherited complement deficiency.	Low complement plus nephritic urine sediment is a nephritis warning. Low C4 in Sjogren can suggest systemic disease/lymphoma risk.
ESR/CRP	ESR often rises with active lupus; CRP may be modest unless infection, serositis, or synovitis.	Very high CRP in lupus should prompt infection evaluation, especially during immunosuppression.

2.3 SLE diagnosis and baseline evaluation

Formal classification criteria are useful for research and pattern recognition, but bedside diagnosis requires a compatible clinical syndrome and exclusion of mimics. Initial evaluation should include CBC with differential, CMP, creatinine, urinalysis with microscopy, urine protein/creatinine ratio, ESR/CRP, ANA, anti-dsDNA, anti-Sm, C3/C4, and selected antibodies based on phenotype. Antiphospholipid antibodies are indicated with thrombosis, pregnancy morbidity, livedo, thrombocytopenia, or stroke-like presentations. Anti-Ro/SSA matters for subacute cutaneous lupus, Sjogren overlap, and pregnancy risk.

Renal screening is mandatory. Lupus nephritis may be discovered before edema or kidney symptoms develop. Concerning findings include proteinuria above 500 mg/day or elevated urine protein/creatinine ratio, hematuria with dysmorphic RBCs, RBC casts, rising creatinine, hypertension, nephrotic syndrome, or low complement with elevated anti-dsDNA. Kidney biopsy is often needed because

urine findings do not reliably identify histologic class or chronicity.

Urine/serology pattern	Likely interpretation	Action
Normal sediment, no proteinuria, stable creatinine	No active nephritis detected now.	Continue periodic monitoring because nephritis can develop later.
Proteinuria above 500 mg/day or elevated protein/creatinine	Possible clinically significant lupus nephritis.	Quantify, repeat, check sediment/complements/dsDNA/BP; refer.
Hematuria plus RBC casts or dysmorphic RBCs	Active glomerulonephritis until proven otherwise.	Urgent nephrology/rheumatology evaluation; biopsy often needed.
Nephrotic-range proteinuria with edema/hypoalbuminemia	Membranous or mixed lupus nephritis possible; thrombosis risk rises.	Renal biopsy and thrombosis assessment.
Rising creatinine with active sediment	Severe nephritis, RPGN, TMA, obstruction, drug toxicity, or infection.	Do not delay urgent evaluation.

2.4 SLE treatment principles

Treatment is severity-driven. Hydroxychloroquine is a foundation drug for most patients unless contraindicated because it reduces flares and helps skin and joint disease; it requires retinal toxicity screening and appropriate dosing. NSAIDs may help short-term arthritis or serositis if kidney function, blood pressure, and GI risk allow. Glucocorticoids are effective for acute flares but should be minimized because chronic exposure causes infection, osteoporosis, diabetes, hypertension, cataracts, mood changes, avascular necrosis, cardiovascular risk, and steroid myopathy.

Steroid-sparing therapy depends on organ involvement. Methotrexate or azathioprine may help persistent arthritis or cutaneous disease. Mycophenolate is central for many lupus nephritis regimens and is also used in some extrarenal disease. Cyclophosphamide is reserved for severe organ-threatening disease such as proliferative nephritis, diffuse alveolar hemorrhage, severe CNS disease, myocarditis, or vasculitic presentations. Belimumab and calcineurin inhibitors are used in selected active SLE and lupus nephritis regimens. Rituximab is commonly used off-label for refractory severe manifestations.

Disease severity	Examples	Therapeutic direction
Mild mucocutaneous/musculoskeletal	Photosensitive rash, oral ulcers, mild inflammatory arthritis, fatigue without organ threat.	Hydroxychloroquine, sun protection, topical therapy, cautious NSAIDs, brief low-dose steroid bridge if needed.
Moderate inflammatory disease	Persistent arthritis, serositis, significant cytopenias, cutaneous disease not controlled with topical therapy.	Hydroxychloroquine plus steroid-sparing drug such as methotrexate, azathioprine, mycophenolate, or belimumab depending on phenotype.
Severe organ-threatening disease	Proliferative nephritis, CNS lupus, pulmonary hemorrhage, myocarditis, severe hemolysis/thrombocytopenia, mesenteric vasculitis.	High-dose glucocorticoids plus organ-directed induction immunosuppression such as mycophenolate or cyclophosphamide; selected biologic or calcineurin therapy.
Pregnancy-associated lupus	Prior nephritis, active disease near conception, anti-Ro/SSA, APS.	Plan pregnancy during quiescent disease; continue pregnancy-compatible hydroxychloroquine; avoid teratogenic agents; coordinate rheumatology and maternal-fetal medicine.

3. Antiphospholipid syndrome

Antiphospholipid syndrome (APS) is an acquired autoimmune thrombophilia defined by thrombosis or characteristic pregnancy morbidity in the presence of persistent antiphospholipid antibodies. It may be primary or associated with SLE. The lupus anticoagulant prolongs phospholipid-dependent coagulation assays in vitro but predicts thrombosis in vivo; the clinical problem is clotting, not bleeding.

Suspect APS in unprovoked venous thrombosis, arterial thrombosis in a young patient, recurrent pregnancy loss, fetal death, severe preeclampsia or placental insufficiency, livedo reticularis, unexplained thrombocytopenia, Libman-Sacks valve lesions, or thrombosis in unusual sites. A single positive test is not enough because antibodies can be transient after infection or inflammation. Persistent positivity requires repeat testing at least 12 weeks apart.

APS test	What it means	High-yield cautions
Lupus anticoagulant	Functional assay showing phospholipid-dependent coagulation interference.	Often strongest thrombosis predictor. Anticoagulants can interfere with testing. Name is misleading because thrombosis risk increases.
Anticardiolipin IgG/IgM	Immunoassay for cardiolipin-associated antibody.	Medium/high titers are more meaningful than low titers; transient positivity occurs.
Anti-beta-2 glycoprotein I IgG/IgM	More specific immunoassay linked to APS.	Triple-positive patients have higher risk than single low-titer positives.

APS test	What it means	High-yield cautions
False-positive RPR/VDRL	Anticardiolipin antibodies react with cardiolipin reagent.	Confirm syphilis status with treponemal testing rather than assuming infection.

Catastrophic APS: Rare but life-threatening. Rapid small-vessel thrombosis causes multiorgan failure, often after infection, surgery, pregnancy, anticoagulation interruption, or lupus flare. Treatment usually combines anticoagulation, high-dose glucocorticoids, plasma exchange and/or IVIG, and treatment of triggers.

Thrombotic APS generally requires long-term anticoagulation. Warfarin remains standard for many high-risk APS patients, especially triple-positive or arterial APS; direct oral anticoagulants are generally avoided in these groups because recurrent event risk is a concern. Obstetric APS is typically treated with low-dose aspirin plus prophylactic heparin during pregnancy, with intensity adjusted for prior thrombosis. Asymptomatic antibody positivity is managed by cardiovascular risk reduction, avoiding estrogen when appropriate, and individualized prophylaxis in high-risk situations rather than automatic indefinite anticoagulation.

4. Sjogren syndrome

Sjogren syndrome is a systemic autoimmune disease centered on lymphocytic infiltration and dysfunction of exocrine glands, especially lacrimal and salivary glands. It may occur as primary disease or secondary to RA, SLE, systemic sclerosis, or another connective tissue disease. Although dry eyes and dry mouth dominate the classic presentation, Sjogren disease can involve lungs, kidneys, peripheral nerves, skin, joints, liver, thyroid, and blood vessels, and it carries increased risk of B-cell non-Hodgkin lymphoma.

Dry eyes cause burning, gritty sensation, redness, photophobia, blurred vision, and keratoconjunctivitis sicca. Dry mouth causes dental caries, oral candidiasis, cracked lips, difficulty swallowing dry foods, frequent water sipping, and altered taste. Parotid enlargement may occur. Extraglandular clues include palpable purpura from vasculitis, interstitial lung disease, distal renal tubular acidosis, interstitial nephritis, peripheral neuropathy, and autoimmune thyroid disease.

Feature	Sjogren interpretation	Clinical action
Dry eyes and dry mouth	Core sicca phenotype from lacrimal and salivary gland dysfunction.	Review medications and mimics: anticholinergics, antihistamines, antidepressants, dehydration, diabetes, radiation, hepatitis C, HIV, IgG4 disease, sarcoid.
Anti-Ro/SSA +/- anti-La/SSB	Supports autoimmune Sjogren and overlap disease.	Anti-Ro is important in pregnancy due to neonatal lupus and congenital heart block risk.
RF positivity and hypergammaglobulinemia	Common B-cell activation pattern.	Do not diagnose RA unless erosive/symmetric inflammatory arthritis pattern fits.
Parotid enlargement, lymphadenopathy, purpura, low C4, cryoglobulins	Systemic disease or lymphoma risk markers.	Evaluate persistent or progressive findings for B-cell lymphoma and vasculitis.
Distal renal tubular acidosis	Tubulointerstitial renal involvement.	Check bicarbonate, potassium, urine pH, nephrolithiasis/nephrocalcinosis risk.
Peripheral neuropathy	Can be small-fiber, sensory neuronopathy, mononeuritis multiplex, or vasculitic.	Neurologic evaluation; severe inflammatory neuropathy may need immunosuppression.

4.1 Sjogren diagnosis and treatment

Diagnosis requires compatible symptoms plus objective evidence of ocular or salivary gland involvement and autoimmune features. Objective eye tests include Schirmer testing and ocular surface staining. Salivary evaluation may include unstimulated salivary flow, salivary gland ultrasound in some practices, or minor salivary gland biopsy showing focal lymphocytic sialadenitis. Biopsy is useful when anti-Ro/SSA is negative or when diagnosis will change management.

Most glandular treatment is symptomatic and preventive: artificial tears, ocular lubricants, topical anti-inflammatory eye drops in selected patients, punctal plugs, hydration, sugar-free lozenges, saliva substitutes, fluoride, dental surveillance, and oral hygiene. Pilocarpine or cevimeline can improve secretions if residual gland function remains but may cause sweating, flushing, urinary frequency, GI upset, bronchospasm risk, or bradycardia concerns. Severe extraglandular disease such as vasculitis, ILD, renal disease, or neuropathy may require systemic immunosuppression.

Lymphoma warning: Persistent unilateral parotid swelling, firm gland enlargement, lymphadenopathy, splenomegaly, palpable purpura, unexplained fevers, night sweats, weight loss, low C4, cryoglobulinemia, or monoclonal gammopathy should prompt lymphoma evaluation.

5. Systemic sclerosis

Systemic sclerosis is a disease of vasculopathy, immune activation, and fibrosis of skin and internal organs. Raynaud phenomenon is often the earliest feature, followed by puffy fingers, skin thickening, digital ulcers, reflux, dysphagia, telangiectasias, calcinosis, interstitial lung disease, pulmonary arterial hypertension, cardiac disease, and renal crisis. Internal organ complications can occur in both limited and diffuse disease, so organ screening is not optional.

Secondary Raynaud is suggested by onset after age 30, severe pain, digital ulcers, pitting scars, asymmetric attacks, abnormal nailfold capillaries, positive ANA, skin thickening, inflammatory arthritis, weakness, dyspnea, or dysphagia. Digital ischemia can lead to ulceration, infection, gangrene, and amputation.

Subtype	Skin pattern	Autoantibody associations	Major risks
Limited cutaneous systemic sclerosis	Skin thickening distal to elbows/knees, often face/neck; long Raynaud history before skin changes.	Anticentromere antibody.	Pulmonary arterial hypertension, digital ischemia, reflux, telangiectasias, calcinosis.
Diffuse cutaneous systemic sclerosis	Rapid proximal skin thickening involving trunk/proximal extremities; shorter Raynaud-to-skin interval.	Anti-topoisomerase I/Scl-70; anti-RNA polymerase III in some.	Interstitial lung disease, renal crisis, tendon friction rubs, cardiac disease, severe skin disease.
Sine scleroderma / overlap	Internal organ disease with minimal skin thickening or mixed features with lupus/myositis/Sjogren.	Variable: U1-RNP, PM/Scl, Ku, others.	Diagnosis may be delayed; screen organs based on phenotype, not skin extent alone.

5.1 Systemic sclerosis organ complications

Gastrointestinal involvement is common. Esophageal dysmotility and lower esophageal sphincter dysfunction cause reflux, dysphagia, esophagitis, strictures, cough, aspiration, and dental erosion. Small bowel dysmotility can cause bacterial overgrowth, bloating, diarrhea, malabsorption, pseudo-obstruction, and weight loss. Treatment is supportive with acid suppression, aspiration precautions, nutrition support, and antibiotics for bacterial overgrowth when present.

Pulmonary disease is a major cause of mortality. Interstitial lung disease presents with exertional dyspnea, cough, crackles, restrictive PFTs, reduced DLCO, and HRCT fibrosis. Pulmonary arterial hypertension presents with progressive dyspnea, fatigue, syncope, right heart strain, loud P2, or isolated DLCO reduction. Screening usually includes PFTs with DLCO, echocardiography, and HRCT when indicated; right heart catheterization confirms PAH.

Scleroderma renal crisis is a hypertensive and kidney emergency. It presents with abrupt severe hypertension, rapidly rising creatinine, headache, visual symptoms, encephalopathy, heart failure, microangiopathic hemolytic anemia, and thrombocytopenia. Risk is higher in early diffuse disease, anti-RNA polymerase III positivity, rapidly progressive skin thickening, tendon friction rubs, and moderate/high-dose glucocorticoid exposure. ACE inhibitors are treatment once renal crisis occurs.

Complication	Clues	Danger
Digital ischemia	Painful Raynaud, ulcers, pitting scars, black discoloration, infection.	Tissue loss, osteomyelitis, gangrene; urgent care.
Interstitial lung disease	Dyspnea, cough, crackles, reduced FVC/DLCO, HRCT fibrosis.	Progressive respiratory failure; common in diffuse disease and anti-Scl-70.
Pulmonary arterial hypertension	Exertional dyspnea, syncope, loud P2, right heart strain, disproportionate low DLCO.	Right heart failure and death; screen even when skin disease is stable.
Renal crisis	Abrupt hypertension, AKI, MAHA, thrombocytopenia, encephalopathy.	Medical emergency; start ACE inhibitor promptly after recognition.
GI dysmotility	Reflux, dysphagia, aspiration, bloating, diarrhea, constipation.	Strictures, aspiration lung disease, malnutrition, pseudo-obstruction.
Cardiac disease	Arrhythmias, conduction disease, myocarditis, pericardial disease, heart failure.	Sudden death risk and heart failure morbidity.

5.2 Treatment

Systemic sclerosis treatment is organ-directed. Raynaud is managed with cold avoidance, smoking cessation, avoidance of vasoconstrictors, calcium channel blockers, PDE-5 inhibitors, topical nitrates, prostacyclin analogs, endothelin receptor antagonists for recurrent ulcers in selected settings, and wound care. Reflux is treated with proton pump inhibitors and aspiration precautions. ILD may be treated with mycophenolate, cyclophosphamide in selected severe disease, rituximab or tocilizumab in selected patients, and antifibrotic therapy for progressive fibrosing disease depending on phenotype. PAH requires specialist-directed pulmonary vasodilator therapy. Glucocorticoids should be used cautiously because of renal crisis risk, especially in diffuse disease.

6. Idiopathic inflammatory myopathies

Idiopathic inflammatory myopathies are systemic autoimmune disorders characterized by muscle inflammation and variable skin, lung, joint, esophageal, cardiac, and malignancy associations. The bedside clue is symmetric proximal muscle weakness, not generalized pain. Patients struggle to climb stairs, rise from a chair, lift objects overhead, comb hair, or get up from the floor. Neck flexor weakness and dysphagia can occur. Myalgia may be present but is not required.

Subtype	Classic clues	Important associations
Dermatomyositis	Proximal weakness plus heliotrope rash, Gottron papules/sign, shawl sign, V-sign, photosensitive poikiloderma, periungual changes.	Malignancy association in adults, ILD in some antibody subsets, calcinosis in juvenile disease.
Polymyositis	Proximal weakness and inflammatory muscle disease without dermatomyositis rash.	Less common as final diagnosis after excluding overlap, necrotizing myopathy, inclusion body myositis, and mimics.
Immune-mediated necrotizing myopathy	Marked CK elevation, severe proximal weakness, necrosis on biopsy with limited inflammation.	Anti-SRP or anti-HMGCR; anti-HMGCR can follow statin exposure and persists after statin cessation.
Antisynthetase syndrome	Myositis plus ILD, inflammatory arthritis, Raynaud, mechanic hands, fever.	Anti-Jo-1 and other antisynthetase antibodies; ILD may dominate prognosis.
Inclusion body myositis	Older adults, slowly progressive weakness, finger flexor and quadriceps involvement, often asymmetric.	Often refractory to immunosuppression; consider when CK is modest and distal weakness is prominent.
Overlap myositis	Myositis with systemic sclerosis, lupus, MCTD, or Sjogren features.	PM/Scl, Ku, U1-RNP; screen for ILD, Raynaud, dysphagia, arthritis.

6.1 Diagnostic approach

The evaluation begins by proving muscle injury and localizing weakness. CK is the most useful screening enzyme; aldolase, AST, ALT, and LDH can also rise from muscle injury. Because AST and ALT are present in muscle, apparent transaminitis in a weak patient may actually be myositis. Myositis-specific and associated antibodies help define phenotype, especially ILD and malignancy risk. MRI can show muscle edema and guide biopsy. EMG supports irritable myopathy. Muscle biopsy is useful when diagnosis is uncertain, inclusion body myositis or necrotizing myopathy is possible, or before prolonged immunosuppression in atypical cases.

Test	What it contributes	Pitfalls
CK	Quantifies muscle injury; often high in active myositis and necrotizing myopathy.	Can be normal in some dermatomyositis or advanced disease; high CK also occurs in rhabdomyolysis, trauma, seizures, drugs, endocrine disease.
Aldolase	May detect muscle involvement when CK is less impressive.	Less specific; interpret with objective weakness.
AST/ALT/LDH	Can rise from muscle injury.	Do not assume liver disease until muscle source considered; check GGT/bilirubin and CK pattern.
Myositis antibody panel	Links phenotype to ILD, malignancy, necrotizing disease, antisynthetase syndrome, or overlap.	Assay quality varies; clinical phenotype remains central.
MRI muscle	Shows edema, inflammation, fatty replacement; guides biopsy site.	Nonspecific after exercise, trauma, or denervation.
EMG	Supports myopathic process and excludes neurogenic mimics.	Does not define antibody phenotype or malignancy risk.
Muscle biopsy	Can show perifascicular atrophy, necrosis, endomysial inflammation, rimmed vacuoles, or overlap features.	Sampling error occurs; choose affected but not end-stage muscle.

6.2 Myositis antibodies and clinical phenotypes

Antibody	Phenotype association	Clinical implication
Anti-Jo-1 and other antisynthetase antibodies	Antisynthetase syndrome: ILD, arthritis, Raynaud, mechanic hands, fever, myositis.	Screen carefully for ILD; lung disease may dominate even when weakness is mild.
Anti-Mi-2	Classic dermatomyositis rash and good treatment response in many patients.	Evaluate skin and muscle; malignancy risk not as high as some other dermatomyositis antibodies.
Anti-MDA5	Clinically amyopathic dermatomyositis and rapidly progressive ILD risk.	Pulmonary symptoms can be an emergency even with little muscle weakness.
Anti-TIF1-gamma	Adult dermatomyositis with increased malignancy association.	Age-appropriate and enhanced malignancy screening depending on risk.
Anti-NXP2	Dermatomyositis; calcinosis in children; malignancy association in adults in some cohorts.	Assess dysphagia, edema, calcinosis, and cancer risk.
Anti-SRP	Immune-mediated necrotizing myopathy, severe weakness, high CK.	Often requires aggressive immunosuppression.
Anti-HMGCR	Immune-mediated necrotizing myopathy, sometimes statin-associated.	Weakness and CK elevation persist despite stopping statin; immunotherapy usually needed.

Antibody	Phenotype association	Clinical implication
PM/Scl, Ku, U1-RNP	Overlap myositis with systemic sclerosis/MCTD/lupus features.	Screen for ILD, Raynaud, sclerodactyly, arthritis, and esophageal disease.

6.3 Treatment and emergencies

Most inflammatory myopathies are treated with glucocorticoids plus early steroid-sparing therapy because long steroid courses cause weakness, infection, osteoporosis, diabetes, and avascular necrosis. Methotrexate or azathioprine may be used for muscle-predominant disease; mycophenolate is common when ILD is present; IVIG is useful in severe dysphagia, refractory dermatomyositis, immune-mediated necrotizing myopathy, or when rapid steroid-sparing is needed; rituximab and other agents are used for refractory disease or selected antibody phenotypes. Inclusion body myositis is largely supportive because response to immunosuppression is poor.

Myositis emergencies: Rapid dyspnea or hypoxemia suggests ILD, aspiration, pulmonary infection, pulmonary embolism, or respiratory muscle weakness. Severe dysphagia risks aspiration and malnutrition. Myocarditis and arrhythmia can occur. Very high CK with renal injury may represent rhabdomyolysis or necrotizing myopathy.

7. Mixed connective tissue disease and overlap syndromes

Mixed connective tissue disease (MCTD) is an overlap syndrome with features of SLE, systemic sclerosis, polymyositis, and rheumatoid-like inflammatory arthritis, classically associated with high-titer anti-U1-RNP antibodies. Findings do not always appear simultaneously. Early disease may present with Raynaud phenomenon, swollen hands, arthralgia, inflammatory arthritis, fatigue, and puffy fingers. Later features include esophageal dysmotility, sclerodactyly, myositis, ILD, pulmonary hypertension, serositis, and mucocutaneous lupus-like findings.

Overlap syndromes are common because immune phenotypes do not respect textbook boundaries. A patient can have systemic sclerosis with myositis antibodies, Sjogren features in SLE, APS in lupus, or RA-like arthritis with Sjogren disease. The practical approach is to list each active organ process, identify antibody clues, and treat the highest-risk manifestation.

Overlap clue	Possible meaning	Why it matters
High anti-U1-RNP with Raynaud and swollen hands	MCTD spectrum.	Pulmonary hypertension and ILD screening are important.
Scleroderma plus CK elevation/proximal weakness	Scleromyositis or overlap myositis.	Treat muscle and screen lung; avoid excessive steroids if renal crisis risk.
SLE plus thrombosis/pregnancy morbidity	Secondary APS.	Anticoagulation strategy may matter more than immunosuppression for thrombosis prevention.
Sjogren plus purpura/low C4/cryoglobulins	Systemic Sjogren with vasculitis/lymphoma risk.	Requires systemic evaluation beyond sicca therapy.
Dermatomyositis plus systemic sclerosis features	Overlap disease with ILD and malignancy considerations.	Antibody profile and organ screening guide treatment.

8. Integrated diagnostic algorithms

8.1 Photosensitive rash, arthralgia, and positive ANA

- Confirm inflammatory features: morning stiffness, objective synovitis, oral/nasal ulcers, photosensitivity, serositis, fever, cytopenias, Raynaud, alopecia, or urinary abnormalities.
- Screen for organ threat: CBC, creatinine, urinalysis with microscopy, urine protein/creatinine ratio, blood pressure, chest symptoms, neurologic symptoms, and pregnancy status when relevant.
- Order targeted serology: anti-dsDNA, anti-Sm, C3/C4, anti-Ro/SSA, anti-La/SSB, and antiphospholipid antibodies when thrombosis, pregnancy morbidity, livedo, or thrombocytopenia is present.
- Exclude mimics: viral infection, drug reaction, dermatomyositis, rosacea, RA, MCTD, systemic sclerosis overlap, malignancy, and medication-induced lupus.
- Treat by severity: hydroxychloroquine foundation for most lupus, steroid bridge for flares, and organ-directed immunosuppression for nephritis/CNS/hematologic/pulmonary/cardiac disease.

8.2 Raynaud phenomenon with systemic symptoms

- Determine primary versus secondary Raynaud. Secondary clues include late onset, severe pain, ulcers, pitting scars, abnormal nailfold capillaries, positive ANA, inflammatory arthritis, skin thickening, weakness, dyspnea, or dysphagia.
- Examine for sclerodactyly, puffy fingers, telangiectasias, calcinosis, digital ulcers, livedo reticularis, purpura, and nailfold capillary dropout.

- Test based on phenotype: ANA, anticentromere, anti-Scl-70, anti-RNA polymerase III, anti-U1-RNP, CK, myositis panel, and antiphospholipid antibodies if thrombosis/livedo is present.
- Screen organs when systemic sclerosis-spectrum disease is possible: PFTs/DLCO, HRCT when indicated, echocardiography, renal function, blood pressure, urinalysis, reflux/dysphagia assessment.
- Treat triggers and vasospasm: cold avoidance, nicotine cessation, avoidance of sympathomimetics, calcium channel blockers first-line for many patients, and escalation for ulcers or ischemia.

8.3 Proximal weakness with high CK

- Confirm true weakness rather than pain-limited effort or fatigue. Test hip flexion, shoulder abduction, neck flexion, chair rise, stair climbing, and dysphagia.
- Quantify injury: CK, aldolase, AST/ALT, LDH, creatinine, urinalysis for myoglobin if rhabdomyolysis is possible.
- Look for phenotype: dermatomyositis rashes, mechanic hands, arthritis, Raynaud, ILD symptoms, statin exposure, malignancy symptoms, and infection risk.
- Order myositis antibodies and assess lungs with PFTs/DLCO and HRCT when dyspnea, cough, crackles, antisynthetase/MDA5 phenotype, or abnormal screening is present.
- Use MRI, EMG, or biopsy when needed; do not delay urgent therapy for rapidly progressive ILD, severe dysphagia, respiratory weakness, or myocarditis.

9. Medication and safety overview

The same medication can help one connective tissue disease and harm another context. Glucocorticoids rapidly control lupus flares and myositis but increase infection risk and may precipitate scleroderma renal crisis at moderate to high doses in susceptible systemic sclerosis patients. Hydroxychloroquine is foundational in lupus but does not treat severe nephritis alone. Mycophenolate maps to lupus nephritis, systemic sclerosis ILD, and myositis-associated ILD, but it is teratogenic and immunosuppressive. Cyclophosphamide can save organs in severe inflammatory disease but carries infertility, cytopenia, infection, bladder toxicity, and malignancy risks.

Drug/class	Common use in this chapter	Major safety issues
Hydroxychloroquine	SLE foundation; cutaneous/joint symptoms; some overlap arthralgia.	Retinal toxicity screening; dose by weight and renal risk; GI upset, rash, rare cardiomyopathy.
Glucocorticoids	Rapid control of SLE flares, myositis induction, severe inflammatory organ disease.	Infection, osteoporosis, diabetes, hypertension, adrenal suppression, avascular necrosis, cataracts, mood changes, steroid myopathy; caution in systemic sclerosis renal crisis risk.
Mycophenolate	Lupus nephritis, systemic sclerosis ILD, myositis-associated ILD.	Teratogenicity, cytopenias, GI effects, infections; monitor CBC/CMP.
Azathioprine	SLE maintenance, selected pregnancy-compatible steroid-sparing, myositis alternatives.	Myelosuppression, hepatotoxicity, infections; TPMT/NUDT15 considerations; interaction with allopurinol/febuxostat.
Methotrexate	Inflammatory arthritis, some cutaneous lupus alternatives, some myositis.	Teratogenic, hepatotoxicity, cytopenias, mucositis, pneumonitis; folate supplementation; monitor CBC/CMP.
Cyclophosphamide	Severe lupus nephritis/CNS/pulmonary hemorrhage, severe ILD or vasculitic disease.	Infertility, cytopenias, infection, hemorrhagic cystitis, bladder cancer, malignancy risk.
Rituximab	Refractory SLE, severe cytopenias, myositis, systemic Sjogren disease, overlap disease.	Infusion reactions, hepatitis B reactivation, hypogammaglobulinemia, infections, impaired vaccine response.
IVIG	Severe dermatomyositis, dysphagia, refractory myositis, some immune cytopenias.	Thrombosis, renal dysfunction, aseptic meningitis, volume load, headache.
Warfarin/heparin	Thrombotic APS and obstetric APS strategies.	Bleeding, monitoring, drug/food interactions; pregnancy considerations differ by agent.

10. High-yield comparison and red flags

Disease	Signature presentation	Signature antibodies	Most dangerous complications
SLE	Photosensitive rash, oral ulcers, nonerosive arthritis, serositis, cytopenias, nephritis, neuropsychiatric disease.	ANA, anti-dsDNA, anti-Sm, anti-Ro/SSA, antiphospholipid antibodies; low C3/C4 during active immune-complex disease.	Lupus nephritis, CNS disease, pulmonary hemorrhage, severe cytopenias, myocarditis, infection from immunosuppression, accelerated cardiovascular disease.

Disease	Signature presentation	Signature antibodies	Most dangerous complications
APS	Venous/arterial thrombosis, pregnancy morbidity, livedo reticularis, thrombocytopenia, valve lesions.	Lupus anticoagulant, anticardiolipin, anti-beta-2 glycoprotein I, persistent on repeat testing.	Recurrent thrombosis, stroke, PE, pregnancy loss, catastrophic APS.
Sjogren	Dry eyes, dry mouth, parotid enlargement, dental caries, arthralgia, fatigue.	Anti-Ro/SSA, anti-La/SSB, ANA, RF, hypergammaglobulinemia.	Lymphoma, vasculitis, ILD, renal tubular acidosis/interstitial nephritis, neuropathy.
Systemic sclerosis	Raynaud, puffy fingers, skin thickening, reflux/dysphagia, telangiectasias, digital ulcers.	Anticentromere, anti-Scl-70/topoisomerase I, anti-RNA polymerase III.	ILD, PAH, renal crisis, digital ischemia, arrhythmia/cardiomyopathy, severe GI dysmotility.
Inflammatory myopathy	Proximal weakness, high CK, rash in dermatomyositis, ILD in antisynthetase/MDA5 disease.	Jo-1, Mi-2, MDA5, TIF1-gamma, NXP2, SRP, HMGCR; overlap antibodies.	Rapid ILD, dysphagia/aspiration, respiratory weakness, myocarditis, malignancy-associated dermatomyositis.
MCTD/overlap	Raynaud, swollen hands, inflammatory arthritis, myositis, esophageal dysmotility, lupus/scleroderma/myositis features.	High anti-U1-RNP; other overlap antibodies possible.	Pulmonary hypertension, ILD, myositis, renal or vascular disease depending on overlap pattern.

Do not miss: nephritis, pulmonary hemorrhage, CNS disease, acute vision loss, digital ischemia, abrupt hypertension with AKI, rapidly progressive dyspnea, severe dysphagia, persistent Sjogren gland enlargement/low C4/purpura, catastrophic APS, and fever in an immunosuppressed patient.

11. Graph-RAG synthesis

The most important graph edges connect antibody, syndrome, organ involvement, emergency, and medication. Anti-dsDNA links to lupus nephritis through immune-complex glomerulonephritis and complement consumption. Anti-Ro/SSA links SLE, Sjogren disease, subacute cutaneous lupus, neonatal lupus, and congenital heart block. Antiphospholipid antibodies link lupus to thrombosis, pregnancy morbidity, livedo reticularis, thrombocytopenia, and false-positive nontreponemal syphilis testing. Anticentromere links limited systemic sclerosis to pulmonary arterial hypertension; anti-Scl-70 links diffuse systemic sclerosis to ILD risk; anti-RNA polymerase III links diffuse disease to renal crisis risk. Myositis antibodies link proximal weakness to ILD, malignancy risk, necrotizing myopathy, or antisynthetase syndrome.

Management edges emphasize severity. Hydroxychloroquine maps to baseline SLE control and flare prevention. Mycophenolate maps to lupus nephritis, systemic sclerosis ILD, and myositis-associated ILD. Cyclophosphamide maps to organ-threatening inflammatory disease. Warfarin and heparin map to thrombotic or obstetric APS. ACE inhibitors map to treatment of scleroderma renal crisis after recognition. Calcium channel blockers and vasodilator escalation map to Raynaud and digital ischemia. Steroids map to acute inflammatory control but also to infection, metabolic complications, osteoporosis, avascular necrosis, and systemic sclerosis renal crisis risk.

Selected references and source coverage notes

Coverage guide: the uploaded rheumatology source pages on systemic lupus erythematosus, drug-induced lupus, systemic sclerosis, Sjogren syndrome, mixed connective tissue disease, and related laboratory marker tables were used to define the chapter scope. The writing is original and synthesized rather than copied.

Fanouriakis A, Kostopoulou M, Andersen J, et al. EULAR recommendations for the management of systemic lupus erythematosus: 2023 update. *Ann Rheum Dis*. 2024.

Merck Manual Professional Edition. Systemic lupus erythematosus, Sjogren syndrome, systemic sclerosis, and idiopathic inflammatory myopathies clinical topic pages. Current online editions.

American College of Rheumatology and EULAR classification/management resources for antiphospholipid syndrome, SLE, systemic sclerosis, and inflammatory myopathies.

Lundberg IE and colleagues. Idiopathic inflammatory myopathy review literature and NIH/PMC summaries on classification, phenotype, and treatment.